SAM THEIL

MECHANICAL ENGINEER AT THE UNIVERSITY OF CINCINNATI

CONTACT

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Cincinnati Ohio

SKILLS

Design

- **Fusion 360**, **Inventor**, used on various projects involving **GD&T**, **FEA Analysis**, designing for CNC and additive manufacturing
- **Autodesk Ultimate CFD**

Mechanical

- **3D Printing** and **Laser Cutting** for rapidly prototype parts
- Manual **CNC machining** for test parts
- **Statistical tolerance analysis** used on tight tolerance parts

Electrical and software

- **Python**, **MATLAB**, **GitHub**
- Front-end and back-end **web development**
- Agile Framework
- **Soldering**
- **Wiring**

EDUCATION

University of Cincinnati

- Completed 57 college credit hours in science & engineering related courses.
- 3.93 GPA

Theil Academy Homeschool

- Will recieve high-school diploma in June 2024
- Weighted GPA: 4.49

PROFESSIONAL EXPERIENCE

eRockets | CAD Designer 2021 - 2023

- Developed a Fusion 360 plugin for automatically designing any model rocket nose cone. Plugin is still in use today.
- Managed 3D print farm operations with other employees; optimized design and tolerances for nose cones.

Contract Web Developer 2023 - Present

- Built web app using low-code tools that utilized OpenAI API (ChatGPT) to automate the writing of user requirements for software development (epics, features, and stories in the Agile framework).
- Successful launch and positive customer response to the web app helped secure $2.5 million funding for my client.

PROJECTS

Rust Airbrakes 2023 - 2024

- An model rocket airbrake system used to make rocket reach specified target altitude each launch

HPR Level 1 rocket 2022 - 2023

- Modeled and built a fully parametric 3D printed rocket. One of the only fully 3D printed high power model rockets made

Scholarship Finder 2024 - Present

- Used Minstral AI model to find scholarships that fit information about the applicant

Dog Feeder 2023 - 2024

- Was bored one weekend - Fully autonomous dog feeder with graphical web interface

TEAMWORK AND LEADERSHIP

American Rocketry Challenge 2020 - 2024

- Built 25+ mid-power model rockets. Launched 150+ mid-power rockets
- Team captain and leader from 2021-2024
  - Used Agile processes to guide team through the design, build, and launch process
  - Implemented SOP for team launch processes
  - Guided and taught new team members in rocketry